

EXAMPLE 6.17

A composite beam is made of wood and reinforced with a steel strap located on its bottom side. It has the cross-sectional area shown in Fig. 6–38a. If the beam is subjected to a bending moment of $M = 2 \text{ kN} \cdot \text{m}$, determine the normal stress at points B and C . Take $E_w = 12 \text{ GPa}$ and $E_{st} = 200 \text{ GPa}$.

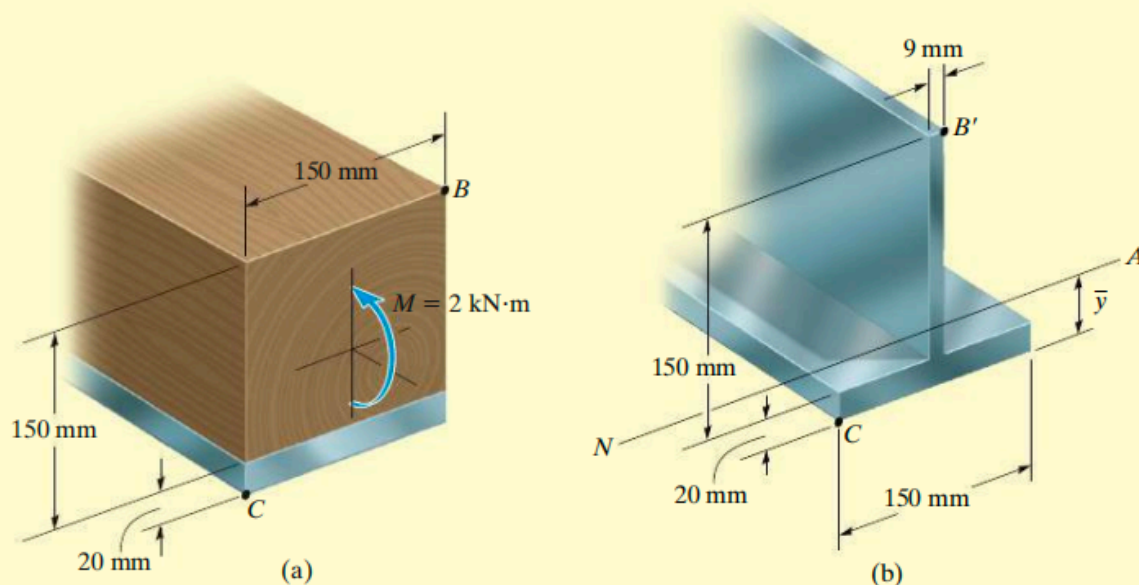


Fig. 6–38

SOLUTION

Section Properties. Although the choice is arbitrary, here we will transform the section into one made entirely of steel. Since steel has a greater stiffness than wood ($E_{st} > E_w$), the width of the wood is *reduced* to an equivalent width for steel. For this to be the case, $n = E_w/E_{st}$, so that

$$b_{st} = nb_w = \frac{12 \text{ GPa}}{200 \text{ GPa}}(150 \text{ mm}) = 9 \text{ mm}$$

The transformed section is shown in Fig. 6–38b.

The location of the centroid (neutral axis), calculated from a reference axis located at the *bottom* of the section, is

$$\bar{y} = \frac{\sum \bar{y}A}{\sum A} = \frac{[0.01 \text{ m}](0.02 \text{ m})(0.150 \text{ m}) + [0.095 \text{ m}](0.009 \text{ m})(0.150 \text{ m})}{0.02 \text{ m}(0.150 \text{ m}) + 0.009 \text{ m}(0.150 \text{ m})} = 0.03638 \text{ m}$$

The moment of inertia about the neutral axis is therefore

$$\begin{aligned} I_{NA} &= \left[\frac{1}{12}(0.150 \text{ m})(0.02 \text{ m})^3 + (0.150 \text{ m})(0.02 \text{ m})(0.03638 \text{ m} - 0.01 \text{ m})^2 \right] \\ &\quad + \left[\frac{1}{12}(0.009 \text{ m})(0.150 \text{ m})^3 + (0.009 \text{ m})(0.150 \text{ m})(0.095 \text{ m} - 0.03638 \text{ m})^2 \right] \\ &= 9.358(10^{-6}) \text{ m}^4 \end{aligned}$$

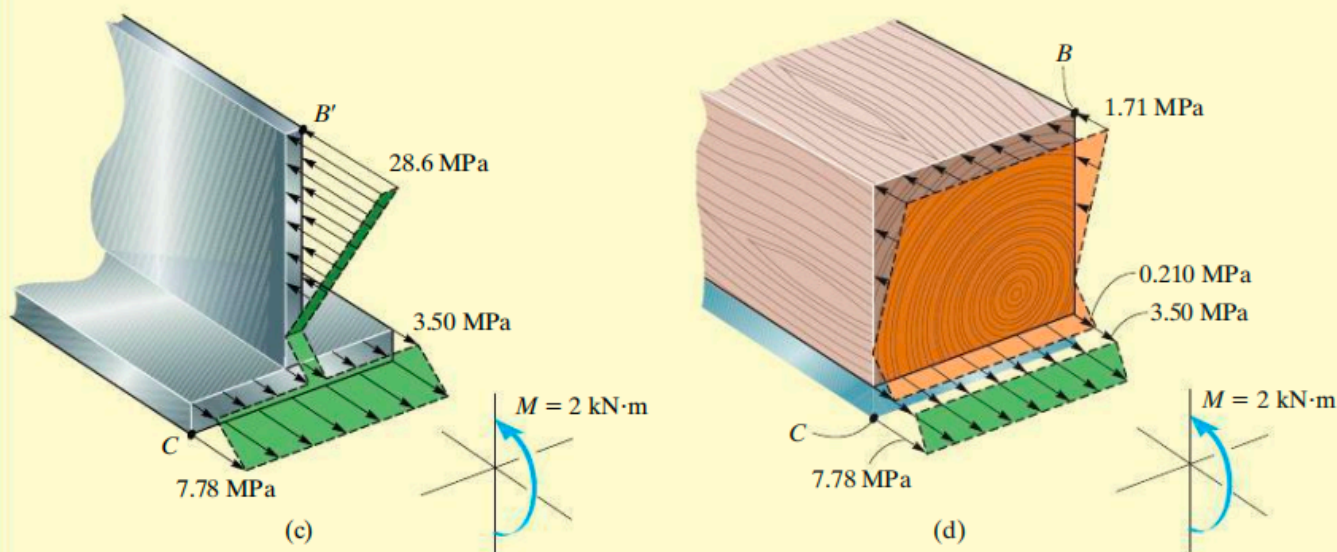


Fig. 6-38 (cont.)

Normal Stress. Applying the flexure formula, the normal stress at B' and C is

$$\sigma_{B'} = \frac{2(10^3) \text{ N}\cdot\text{m}(0.170 \text{ m} - 0.03638 \text{ m})}{9.358(10^{-6}) \text{ m}^4} = 28.6 \text{ MPa}$$

$$\sigma_C = \frac{2(10^3) \text{ N}\cdot\text{m}(0.03638 \text{ m})}{9.358(10^{-6}) \text{ m}^4} = 7.78 \text{ MPa} \quad \text{Ans.}$$

The normal-stress distribution on the transformed (all steel) section is shown in Fig. 6-38c.

The normal stress in the wood at B in Fig. 6-38a, is determined from Eq. 6-21; that is,

$$\sigma_B = n\sigma_{B'} = \frac{12 \text{ GPa}}{200 \text{ GPa}}(28.56 \text{ MPa}) = 1.71 \text{ MPa} \quad \text{Ans.}$$

Using these concepts, show that the normal stress in the steel and the wood at the point where they are in contact is $\sigma_{st} = 3.50 \text{ MPa}$ and $\sigma_w = 0.210 \text{ MPa}$, respectively. The normal-stress distribution in the actual beam is shown in Fig. 6-38d.

A composite beam is made of wood and reinforced with a steel strap located on its bottom side. It has the cross-sectional area shown in Fig. 6-38a. If the beam is subjected to a bending moment of $M = 2 \text{ kN} \cdot \text{m}$, determine the normal stress at points B and C. Take $E_w = 12 \text{ GPa}$ and $E_{st} = 200 \text{ GPa}$.

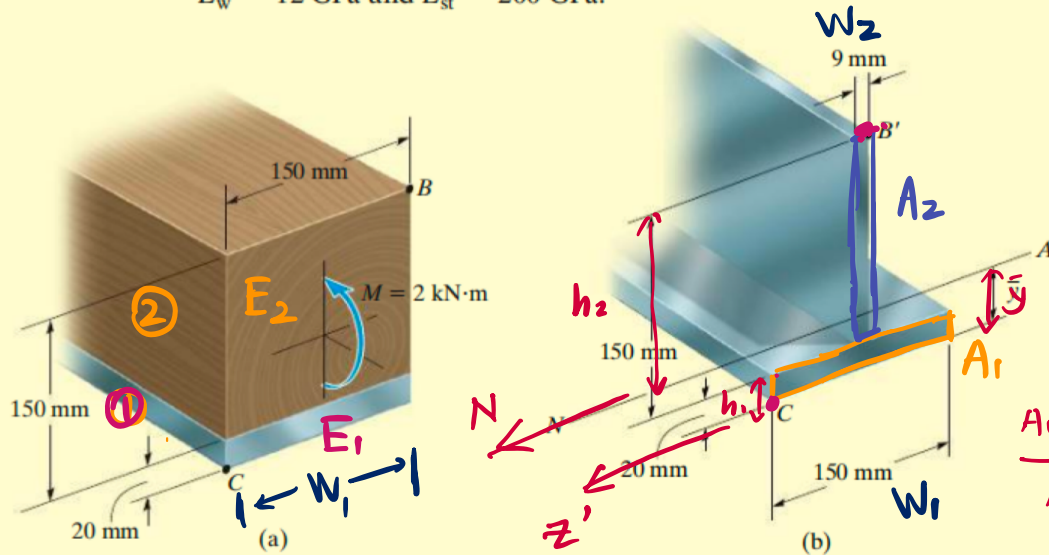


Fig. 6-38

$$y_B = h_1 + h_2$$

$$y_C = 0.$$

$$\frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \bar{y}$$

$$W_1 = 150 \text{ mm}$$

$$\text{等效: } W \propto E$$

$$n = \frac{E_1}{E_2} = \frac{E_2}{E_1}$$

$$n < 1$$

$$W_2 = n W_1 \Rightarrow W_2 = 9 \text{ mm}$$

```

w1=150e-3
Mz=2e3
E2=12e9
E1=200e9

h1=20e-3
h2=150e-3

w2=w1/E1+E2

A1=w1*h1
A2=w2*h2

y1=h1/2
y2=h1+h2/2

ybar=(A1*y1+A2*y2)/(A1+A2)

I1c=1/12*w1*h1**3
I2c=1/12*w2*h2**3

d1=y1-ybar
d2=y2-ybar

I1=I1c+A1*d1**2
I2=I2c+A2*d2**2

Iz=I1+I2

cB=h1+h2-ybar
cC=ybar

sigmaBp=Mz*cB/Iz
sigmaC=Mz*cC/Iz
    
```

$$I = I_c + A d^2$$

$\frac{1}{2} b h^3$

$C \rightarrow NA$

$$\sigma_{B'} = \frac{M_z c_B}{I_z} = 28.56 \text{ MPa}$$

$$\sigma_C = 7.78 \text{ MPa}$$

$$\sigma_B = \sigma_{B'} n$$

